

SUMMARY

Tanmoy is an AI and Machine Learning (ML) Consultant, Coach, Research Scholar, Experienced Programmer and Freelance Research Guide. He perceived that research is incomplete without practical implementation based on his diversified experience in the industry as well as in academia and

Presently he is a Senior Data-Scientist where his primary responsibility is **AI and ML** solutioning and team building. He also guides researchers of AI and ML fields and provides training to aspiring AI and ML practitioners. His work area encompasses *speech processing, medical data processing, computer vision, natural language processing, medical imaging, and IoT-Edge computing*.

From 2016 to 2021 he worked full-time at the **University of Johannesburg** on fully funded Global Excellence Stature (GES) positions — as a **Doctoral Research Fellow** (Jul 2016 – Jun 2019), completing his PhD within the stipulated three years, and then as a **Post-Doctoral Research Fellow** (Oct 2019 – Sep 2021).

Tanmoy started his professional career in **SIEMENS** as a programmer and provided niche solutions to problems in wide-ranging projects encompassing telecommunication, electronic vehicles and smart-grid billing. His programming skills in different programming languages like C, C++, Java, SQL and his expertise in system design flourished during this tenure. He had the opportunity to be part of many complete project development life cycles and learn the craft.

EXPERIENCE

DeliverHealth Solutions

REMOTE, INDIA

[JULY '24 – PRESENT]

Senior Data Scientist

Led research and development for AI-driven product and solution ideation, driving organization-wide AI adoption initiatives.

Collaborated with cross-functional stakeholders, including Product Owners and partners such as Google and Microsoft, to design and deploy scalable AI solutions addressing critical business challenges.

Medical Coding Co-pilot: System to assist human coders so that the time to code a single document can be significantly reduced. Here, we are leveraging proprietary historical data to finetune a Small Language Model in a continuous learning loop.

Automatic Medical Note Generation: Currently building a self service platform for Automatic Medical Note Generation for Physicians using LLM (Gemini 3.1 Pro). Its a template based generation leveraging proprietary historical data.

RAG-based Medical Correction Engine: This is a RAG based solution to automatically correct ASR drafts before they are delivered. Reduced significant manual correction efforts and improved document quality.

Finetuning STT Models with Proprietary Data: In this engagement our goal was to create our own Med-ASR engine leveraging DHS proprietary data. Finetuned models like Gemini-1.5-Pro, Whisper and went on to reach 12% Medical WER.

Cloudcraftz Solutions

KOLKATA, INDIA

[NOV '22 – JUNE '24]

Senior Data Scientist

Primary responsibility was to lead and mentor a young team having diversified skills in Data Science, Quantitative Analytics and Software Engineering, and deliver complex solutions primarily for the Finance domain.

Option Trading Strategy Backtest and Analytics Solution: This Option Trading Backtest system is being used to test complex option trading strategies. The primary skills required for this project are mix of Software Engineering (SE), Financial Analytics and Data Science.

Explainable AI: This product aimed to enhance transparency and interpretability in machine

learning models using concepts like SHAP values and accumulated Local Effect (ALE) to visualize different aspects of ML models. Was involved in developing this Explainable AI product.

Data Visualization Platform: This is a highly intuitive system providing in-depth data analysis with minimum human intervention. Lead this platform development project.

Veersa Technologies

KOLKATA, INDIA

[MAY '22 – OCT '22]

Data Scientist (Consulting)

Primary responsibility was ML model building with both tabular and image data. Other responsibilities involve data engineering, ML model productionizing, and ML team building. Executed projects in the healthcare domain.

Insurance Claim Denial Prediction: The primary target of this project is to predict whether claims submitted by the health institute will get denied or approved. Even as further analysis the reasons behind the denial are also depicted. Reducing the number of denials can have a great impact on the revenue of the health institute. Data is extracted in tabular form with relevant features and after required pre-processing that data is used as training data for the XGBoost classifier. Model accuracy was 89% although there was further scope for improvement.

Patient Data De-Duplication: Broken chain of information is a challenge for healthcare providers because this affects service quality and consumes storage space. The **record linkage** concept is used to solve this problem.

Disease Detection from X-Ray images: Detecting diseases from an X-ray can lead to a significant reduction of time and manual intervention. Used ResNet34 model to achieve 84% accuracy.

Clinical Trial Enrollment Prediction: This was a proof of concept on the publicly available high-quality data where the target was to predict the number of people finally enrolled for a clinical trial. The prediction accuracy achieved was 75% but there was further scope for improvement.

Handwritten Medical Form Reader: Digitization of handwritten medical forms is a big challenge for health service providers mainly due to varying hand-writing styles. AWS Textract is used to extract handwritten texts with high accuracy and the solution is built around the Textract output.

University of Johannesburg

JOHANNESBURG, SOUTH AFRICA

[JUL '16 – SEP '21]

Post Doctoral Research Fellow

[OCT '19 – SEP '21]

Full-time funded research position under the Global Excellence Stature (GES) fellowship. Built and evaluated deep learning models for speech emotion recognition, authored survey work on emotion recognition in the context of the 4th-industrial revolution, and initiated two international research collaborations — with King Saud University (Riyadh, Saudi Arabia) and Birla Institute of Technology and Science (Pilani, India).

Doctoral Research Fellow

[JUL '16 – JUN '19]

Full-time funded doctoral position under the GES scholarship. Completed the doctoral thesis "Detecting emotions from speech using Machine Learning Techniques" within the stipulated three years, proposing a novel feature set for speech emotion recognition (SEGRT) and a new algorithm for precise speech endpoint detection.

SIEMENS

KOLKATA, INDIA

[Jan '07 – Sep '15]

Associate Consultant

[Jan '13 – Sep '15]

- Designed and implemented a Telecom Service Provisioning system based on a RedHat Linux multi-tiered application using Java Spring Framework. Technologies used *Java Spring, Oracle*.
- Customization of Telecom Service Provisioning system using *Java, SOAP, XML, Oracle* on *Linux* platform.
- Designed and implemented CRM Service Provisioning integration module for GSM-R Billing system using *Java, Oracle PL/SQL*.
- Designed and implemented the Billing module for GSM-R Billing system using *C/C++, Oracle PL/SQL* on *Linux* platform.

Sr. System Engineer

[Jan '11 – Dec '12]

- Designed and implemented a Charging module for the Electronic Car billing system using C/C++ and Oracle.
- Designed and implemented GUI based Telecom Call Analysis system. Features include Call tracking, Zone based call analysis with Graphical simulations. Used C++ based Qt GUI Library.

System Engineer

[Jan '09 – Dec '10]

- Designed and implemented Bill Printing and Bill By Email module for a Telecom Billing system. **Technologies:** Java, ProC/C++, Oracle PL/SQL.
- Designed and implemented a Charging module for a Telecom Interconnect Billing system using Java and Oracle. **Technologies:** Java, Oracle.

Associate System Engineer

[Jan '07 – Dec '08]

- Single handedly provided System Support for IVR and USSD Solution. Technologies used: Java, C++, Linux, awk script, shell script.
- Developed an automated monitoring system using Java, shell script, awk script to improve client experience.

TECHNICAL EXPERTISE

- **Programming Languages:** Python, C/C++, R, Matlab, Java, JavaScript, SQL, Scala, Julia.
- **Libraries:** Few prominent python libraries are: ollama, langchain, langgraph, unsloth, pytorch, pandas, matplotlib, scipy, opencv, pyspark.
- **Technologies/Platforms:** Cloud (AWS, GCP), Linux, Docker container, MLOps (KubeFlow), REST, SOAP, XML.
- **Databases:** RDBMS (Oracle, PostgreSQL, SQL Server, MySQL), NoSQL (MongoDB, DynamoDB, Cassandra), Cloud DB (Amazon RDS).

EDUCATION

Doctor of Philosophy

2019, UNIVERSITY OF JOHANNESBURG

JOHANNESBURG, SOUTH AFRICA

Thesis Title: Detecting Emotions From Speech Using Machine Learning Techniques.

Area of research: Applied Machine Learning and Deep Learning, Speech Signal Processing.

Master of Science

2013, LOVELY PROFESSIONAL UNIVERSITY

KOLKATA, INDIA

Specialization: Information Technology.

Subjects: Computer Organization, Programming(C++/Java), Data Structure, Computer Graphics, Artificial Intelligence, System Software.

Post-Graduate Diploma in Computer Applications

2004, DOEACC SOCIETY, MINISTRY OF IT, GOVT. OF INDIA.

KOLKATA, INDIA

Advanced Diploma in Computer Applications (A-Level).

Bachelor of Science

2002, UNIVERSITY OF CALCUTTA

KOLKATA, INDIA

Economics(Hons), Computer Science, Mathematics

CERTIFICATES

1. **Deep Learning with PyTorch : Neural Style Transfer**
Coursera
2. **Neural Network from Scratch in TensorFlow**
Coursera

3. **Billing and Revenue Management**
Oracle University
 4. **Diploma in Computer Networking**
CMC Limited
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REFERENCES

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